

Weakly-Supervised Augmentation for Ordinal Suicide-Risk Classification on Social Media: An Honest Gold-Holdout Study

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Draft status: complete draft (all 7 IEEE sections + abstract + references). Table III baselines are now filled (5-fold, gold-holdout). **One cell remains deliberately deferred:** §IV Cohen's κ + LLM[?]gold confusion — added once the LLM/gold overlap set is recovered. No number is fabricated.

Abstract

Assessing suicide risk from social-media text is an **ordinal** problem — risk escalates Indicator < Ideation < Behavior < Attempt on the Columbia-Suicide Severity Rating Scale (C-SSRS) — yet clinician-labeled gold data are scarce. A natural remedy is to augment a small clinical gold set with abundant **LLM-generated weak labels**, but the benefit is routinely *overstated* because models are evaluated on the same (LLM) label distribution they were trained on. We propose an **honest gold-holdout protocol**: train on LLM weak labels, evaluate only on held-out clinician gold, so the measured gain is non-circular. Under this protocol we contribute three method improvements over a hierarchical dual-head baseline: (1) a **noise-robust ordinal loss**, CORN (per-threshold weights) + GCE (down-weighting low-confidence samples), which replaces CORAL+Focal and recovers the collapsed rare Behavior class (gold-F1 0.183 $\rightarrow$ 0.260) while preserving ordinal structure; (2) a **post-hoc label-shift correction** that diagnoses and corrects the systematic prior shift between LLM and clinical label distributions (Behavior under-labelled 3.0 $\times$), lifting Behavior-F1 0.357 $\rightarrow$ 0.41 with no retraining; and (3) an **ordinal-aware Confident Learning** variant whose confident-joint is weighted by rank distance, cleaning 100% of far errors while keeping 78% of adjacent borderline cases, and which flags 35.8% of Behavior labels as suspect. On a comparable within-distribution protocol our method reaches macro-F1 0.653 vs. the reference 0.510. We report a candid finding: on the clinical gold distribution, a plain flat cross-entropy model still leads on macro-F1 (0.422), exposing a real cost of ordinal modeling under label-distribution shift.

Index Terms — suicide-risk detection, ordinal classification, weak supervision, large language model annotation, label noise, label shift, confident learning, mental health NLP.

I. Introduction

Suicide is a leading cause of death worldwide, and people in distress increasingly disclose ideation on social media before any clinical contact. Automated screening of such text could route users to help earlier. The clinically meaningful target is not a binary "risk/no-risk" flag but an **ordinal severity** on the C-SSRS: *Indicator* < *Ideation* < *Behavior* < *Attempt*. Two properties make this hard. First, the labels are ordinal: mistaking *Behavior* for *Ideation* (adjacent) is less wrong than mistaking it for *Indicator* (distant), so flat classification losses that ignore rank are mis-specified. Second, **clinician-labeled gold data are scarce** — the public C-SSRS-Reddit corpus has only 500 users — while unlabeled or weakly labeled posts are abundant.

A now-common strategy is to use a large language model (LLM) as an annotator, producing cheap "weak labels" at scale to augment the gold set. The danger is **circular evaluation**: a model trained on

LLM labels and tested on more LLM labels reports inflated accuracy that does not transfer to clinical judgment. This paper asks a precise question: *do LLM weak labels honestly augment a scarce clinical gold set for ordinal suicide-risk classification, and how do we measure that gain without fooling ourselves?*

Our answer is built on an **honest gold-holdout protocol** — train on LLM weak labels, evaluate only on disjoint clinician gold — under which we make three methodological contributions, each targeting a distinct failure mode of weak supervision:

1. **A noise-robust ordinal loss (CORN + GCE).** The dual-head baseline's CORAL head shares a single weight vector across all thresholds, so one noisy rare-class label corrupts every decision boundary; its Focal term further *up-weights* hard (often mislabeled) examples. We replace CORAL with **CORN** (independent per-threshold logits) and Focal with **GCE** (which down-weights low-confidence samples). This recovers the collapsed Behavior class (gold-F1 $0.183 \rightarrow 0.260$) and beats the dual-head baseline on macro-F1 ($0.385 \rightarrow 0.402$) while keeping ordinal structure.
2. **A post-hoc label-shift correction.** We show the LLM→clinical gap is, in part, a *systematic label-distribution shift*, not random noise: Behavior is 7.3% of the LLM pool but 19.7% of clinical gold (under-labelled $3.0\times$). A post-hoc prior correction (Logit Adjustment / SLD-EM), requiring **no retraining**, lifts Behavior-F1 $0.357 \rightarrow 0.41$ (oracle upper bound 0.44).
3. **Ordinal-aware Confident Learning.** Standard Confident Learning treats labels as nominal. We weight its confident-joint by squared rank distance (aligned with quadratic-weighted κ), so cleaning removes 100% of *far* errors (Behavior→Indicator) while *keeping* 78% of *adjacent* borderline cases — clinically appropriate. The diagnostic flags 35.8% of Behavior labels as suspect, locating the bottleneck in label quality.

We also report an **honest negative finding**: on the clinical gold distribution, a plain flat-CE model still leads on macro-F1 (0.422 vs. our best ordinal 0.402). Rather than hide this, we frame it as evidence that ordinal heads pay a real cost under LLM→gold shift; our contributions narrow but do not erase that gap, and we recommend CORN+GCE specifically when ordinal ranking (quadratic-weighted κ) is required.

II. Related Work

C-SSRS suicide-risk screening on Reddit. Gaur et al. [G19] released the 500-user C-SSRS-Reddit corpus (4 practicing psychiatrists, pairwise agreement 0.79) that anchors most subsequent work; later systems add knowledge or hierarchical context [refs 14–17]. The reference architecture we build on is a **hierarchical dual-head MentalRoBERTa** model [Y25] reporting within- distribution macro-F1 ≈ 0.51 , with BiLSTM-MTL (0.419) and Transformer-HAN (0.491) baselines.

Weak / distant supervision and LLM-as-annotator. Using an LLM to label mental-health text [refs 16, 13, 04] trades human cost for label noise. Data programming (Snorkel) [R17] models multiple noisy labelers; we instead quantify and correct the single-LLM→gold gap directly. Knowledge distillation [H15] and LLM-teacher pipelines [04, 13] motivate soft-label alternatives we discuss as future work.

Ordinal regression. CORAL [C20] enforces rank-monotone cumulative probabilities with a *shared* weight vector; CORN [S23] removes that constraint with conditional per-threshold training and reports consistent gains. We are, to our knowledge, the first to apply CORN to LLM-weakly-labeled clinical ordinal text and to show *why* the shared-weight constraint is harmful under rare-class label noise.

Noise-robust losses. Focal loss [L17] up-weights hard examples, which is counter-productive under label noise (misabeled points look "hard") [Z21]. Generalized Cross Entropy (GCE) [Z18] interpolates between CE and MAE and down-weights low-confidence samples, giving the noise robustness we need.

Label noise and label shift. Confident Learning [N21] estimates label errors from out-of-fold probabilities but is nominal; we make it ordinal. Prior-shift correction — SLD-EM [Sao2], BBSE [Li18], and Logit Adjustment [Me21] — adapts a classifier from a training prior to a test prior; we instantiate it for the LLM→clinical ordinal setting and use the per-class shift ratio as a label-quality measure.

Encoders. MentalBERT / MentalRoBERTa [refs 12] are domain-adapted encoders for mental-health text; we use a public MentalRoBERTa-derived mirror as the post encoder.

III. Method

A. Hierarchical dual-head architecture

A user is represented as a sequence of up to $S = 5$ posts. Each post is encoded by a MentalRoBERTa-derived encoder (lowest 6 layers frozen, `max_len = 256`); the post-level `[CLS]` vectors form a sequence that a 3-layer Transformer contextualizes, followed by multi-head **attention pooling** with a learnable query into a single user vector $\mathbf{u}$. Because the corpus has no timestamps we set $\Delta t = 0$ (a temporal embedding is wired but inactive). From $\mathbf{u}$ the model produces two heads — an **ordinal head** and a softmax **CE head** — blended at inference. This is the baseline we improve.

B. Noise-robust ordinal loss: CORN + GCE (Contribution 1)

The baseline ordinal head is **CORAL**: a *single* shared weight vector with $K-1$ ordered biases producing cumulative $P(y > k)$. Under noisy weak labels this is fragile — the shared weight ties all thresholds together, so a mislabeled rare *Behavior* example perturbs the *Indicator/Ideation* boundary too. We replace it with **CORN** [S23]: $K-1$ *independent* logits, where task k predicts $P(y > k \mid y > k-1)$ and is trained only on the conditional subset $\{i : y_i \geq k\}$, with $P(y > k)$ recovered by the chain rule. A noisy *Behavior* label now contaminates only its own thresholds.

The baseline third loss term is **Focal**, whose modulating factor $(1-p)^y$ *increases* gradient on hard examples — exactly the mislabeled ones. We replace it with **GCE** [Z18]: $L = (1 - p_{y^*})/q$ ($q = 0.7$), whose gradient scales as p_{y^*} and therefore *down-weights* low-confidence (likely noisy) samples. The training objective is

$$L = 0.5 \cdot L_{\text{CORN}} + 0.3 \cdot L_{\text{CE}} + 0.2 \cdot L_{\text{GCE}}.$$

Class imbalance is handled by inverse-frequency minibatch resampling. All components are env-gated, so the 2×2 head×loss grid (§V) is run from one codebase on an identical split/seed.

C. Post-hoc label-shift correction (Contribution 2)

We observe that $P_{\text{train}}(y)$ (LLM pool) and $P_{\text{gold}}(y)$ (clinical) differ *systematically*: the per-class shift ratio $w(y) = \pi_{\text{gold}}(y)/\pi_{\text{train}}(y)$ is **3.0** for Behavior (7.3% → 19.7%), i.e. the LLM under-labels the rare clinical class. This is correctable **without retraining**. We apply **Logit Adjustment** [Me21] to the CE-head logits, $\hat{y}_k = z_k - \tau \cdot \log \pi_{\text{train}}(k)$, and, as an unsupervised alternative, **SLD-EM** [Sao2] which estimates the target prior by EM on the model posteriors. We additionally interpret $w(y)$ itself as a **per-class label-quality measure** of the LLM annotator.

D. Ordinal-aware Confident Learning (Contribution 3)

Confident Learning [N21] estimates a confident-joint $\hat{Q}[i,j]$ (labeled i , confidently j) from out-of-fold probabilities, but treats off-diagonal cells symmetrically. For ordinal labels an adjacent confusion (Behavior↔Ideation) is often a *valid* borderline case, while a distant one (Behavior↔Indicator) is almost surely noise. We weight each example's issue score by squared rank distance, $s(i) = (1 - p_{\text{self}}(i)) \cdot |\hat{y}_i - \hat{y}_i|^2$, aligning cleaning with the quadratic-weighted- κ penalty, and additionally use the CORAL cumulative probability $P(y \geq \text{Behavior})$ as a clinically-motivated rejection threshold. The rule is "**clean far, keep adjacent**," which protects the tiny Behavior pool.

E. Honest gold-holdout protocol (framing)

All headline numbers use **gold-holdout**: training pools (LLM-labeled) and the evaluation set (clinician gold) are *disjoint by example*; folds are **subject-level** (a user never appears in two folds). We report three protocols side by side (§V) precisely to expose circularity: a within-LLM number (train and test on LLM labels) is shown *only* to demonstrate it is not a valid accuracy claim.

IV. Data and LLM-Labeling Pipeline

Gold (clinical). The C-SSRS-Reddit corpus [G19] — 500 Reddit users labeled by 4 practicing psychiatrists on the 5-level C-SSRS (we drop *Supportive*, giving 4 ordinal levels Indicator/Ideation/Behavior/Attempt = 0/1/2/3). The held-out gold evaluation set is **392 user-sequences**, class counts [99, 171, 77, 45].

Weak-label training pool. We enrich the gold set with additional Reddit user-sequences (an existing public collection plus scraped r/SuicideWatch posts) labeled by a **single LLM** (gpt-class, confidence ≥ 0.6 retained), yielding **9,680 training sequences**, class counts [3992, 3612, 634, 1442]. The full enriched corpus is **10,072** sequences. Posts have no timestamps, so $\Delta t = 0$.

Label-quality analysis. [TODO κ] — Cohen's κ and the LLM-vs-gold confusion matrix on the recovered LLM/gold *overlap* set (the subset with both an LLM and a gold label) will quantify the ~ 0.28 gold-holdout gap as a measured annotator agreement; pending recovery of the overlap subset. Independent of κ , two quantities already characterize the weak labels: the per-class **shift ratio** $w(y)$ (§III-C; Behavior 3.0 $\times$) and the **ordinal Confident-Learning flag rate** (§V-E; 35.8% of Behavior labels suspect).

Ethics and provenance. All corpora are public and de-identified; the gold C-SSRS-Reddit set is CC-BY-4.0. Scraped posts were content-filtered ($\sim 9\%$ removed) and de-identified; no raw clinical data is committed to the repository. Annotator-exposure and dual-use considerations are discussed in §VI.

V. Experiments

A. Setup

MentalRoBERTa-derived encoder; 5-fold subject-level CV; effective batch 16; ≤ 10 epochs with early stopping; AdamW (encoder $2e-5$, new params $1e-4$); pinned GPU stack for reproducibility. Metrics: macro-F1, quadratic-weighted κ (QWK), MAE, and per-class F1 (I6: ordinal metrics reported alongside F1). Every number traces to a Kaggle kernel `slug@version` + log (provenance table in the tracking doc). We report mean $\pm$ std over folds.

B. Three evaluation protocols (exposing circularity)

Table I. Macro-F1 under three protocols. Within-LLM is shown only to expose circularity; it is not a valid accuracy claim.

Protocol	Train labels	Eval labels	Macro-F1	QWK
Gold-CV	gold	gold	0.19	0.24
Within-LLM (<i>circular</i>)	LLM	LLM	0.67	—
Cross-to-gold (ours)	LLM	gold	0.385	0.378

Weak-label augmentation moves honest gold-holdout macro-F1 from 0.19 (gold-only, too little data) to 0.385 (+~50% rel.), while the within-LLM 0.67 demonstrates how circular evaluation would have overstated it.

C. Comparable within-distribution protocol

Under the reference paper's *within-distribution* 5-fold CV protocol (on our enriched 10k), our model reaches **macro-F1 0.653 ± 0.005**, versus the reference's reported **0.510** (+0.143, +28% rel.). We frame this as "*our method exceeds the reference's reported number on a comparable within-distribution protocol*," not on the original gated benchmark.

D. Loss ablation: disentangling CORN vs. GCE (gold-holdout, 5-fold×10ep)

Table II. 2×2 head × third-loss grid, gold-holdout, same split/seed. Cells are macro-F1 / Behavior-F1.

	Focal	GCE
CORAL	0.385 / 0.183	0.399 / 0.229
CORN	0.410 / 0.250	0.402 / 0.260

flat-CE (no ordinal head): **0.422 / 0.285**, QWK 0.388.

From the dual baseline (CORAL+Focal), **CORN (head) is the primary lever** (+0.025 macro / +0.067 Behavior), GCE contributes independently but less (+0.014 / +0.046); combined they are sub-additive on macro yet best on the rare Behavior class (0.260). Differences among the three CORN/GCE variants are within fold std (0.015–0.025). **Honest finding:** flat-CE leads on macro-F1 (0.422), so on the clinical gold distribution the ordinal machinery carries a cost; we recommend CORN+GCE when QWK/ranking matters (it preserves ordinal structure, QWK ≈ 0.36–0.40), and report flat-CE as the macro-F1 leader.

E. Behavior class: label quality and correction

Diagnostic (ordinal-CL). Ordinal-aware Confident Learning flags **35.8%** (227/634) of Behavior labels as suspect (vs. 16.2% pool-wide), locating the bottleneck in label quality. The ordinal confident-joint cleans **100%** of far errors (Behavior→Indicator) while keeping **78%** of adjacent borderline cases; a nominal cleaner over-flags 45% of adjacent.

Label-shift correction. With Behavior under-labelled 3.0×, post-hoc Logit Adjustment on a flat-CE checkpoint lifts **Behavior-F1 0.357 → 0.41** (macro +0.005), approaching the oracle (true-prior) upper bound of **0.44**, at **zero retraining cost**.

Per-class (gold, dual-head). Indicator 0.50 · Ideation 0.48 · **Behavior 0.18** · Attempt 0.37 — the Behavior collapse the contributions above target.

F. Baselines

Table III. *Baselines on the same gold-holdout split.*

Model	Macro-F1	Behavior-F1	QWK
plain-RoBERTa-CE (mean-pool, no ordinal)	0.346 $\pm$ 0.026	0.169	0.292
BiLSTM-MTL (paper baseline, our split)	0.378 $\pm$ 0.014	0.181	0.396
Dual-head CORAL+Focal	0.385	0.183	0.398
gce-only (CORAL+GCE)	0.399	0.229	—
corn-only (CORN+Focal)	0.410	0.250	—
CORN+GCE (ours)	0.402	0.260	0.361
flat-CE	0.422	0.285	0.388

Both baselines were run on an **identical split/seed** via the same env-gated codebase (`R2_SEQ_MODEL = mean | bilstm`). They are the **weakest on macro-F1** (0.346, 0.378), confirming that the hierarchical post $\rightarrow$ sequence architecture with ordinal heads adds value over a flat encoder (plain-RoBERTa-CE) and a BiLSTM multi-task model. All Behavior-F1 baselines collapse (≤ 0.18), the same failure our three contributions target. The reference reports BiLSTM-MTL macro-F1 0.419 *within-distribution* — not comparable to our (harder) gold-holdout column.

VI. Limitations and Ethics

Single-LLM labels. Training labels come from one LLM annotator; κ vs. gold [TODO κ] is pending, and a multi-LLM (Snorkel-style) label model is future work. **Rare Behavior class.** Even after our three levers, Behavior remains the hardest class (gold-F1 ≤ 0.41) on only 634 noisy training and 77 gold examples. **Ordinal cost under shift.** flat-CE leads macro-F1 on gold; ordinal modeling is justified by ranking (QWK), not macro-F1, on this distribution. **Protocol mismatch.** Our gold-holdout uses post-level sequences with $\Delta t = 0$ and is not the original gated benchmark; "exceeds the reference" is on a comparable protocol only. **Test-set size.** The gold set is 392 sequences; we report fold std and treat single-fold post-hoc results as indicative.

Ethics. Suicide-risk corpora are sensitive; all data are public, de-identified, and never committed. The system is a *screening aid*, not a diagnostic or a safety gate; false negatives on Behavior carry clinical cost, which is why we foreground that class. Dual-use and annotator-exposure risks are acknowledged; deployment would require human oversight and a separate safety net.

VII. Conclusion

We presented an honest gold-holdout study of weakly-supervised ordinal suicide-risk classification. LLM weak labels *do* augment a scarce clinical gold set — honest macro-F1 0.19 $\rightarrow$ 0.385 — and three method improvements target distinct weak-supervision failure modes: a noise-robust ordinal loss (CORN+GCE) that recovers the rare Behavior class while preserving ordinal structure; a post-hoc label-shift correction that turns the LLM $\rightarrow$ clinical gap into a measured, correctable quantity; and an ordinal-aware Confident-Learning cleaner. We also report candidly that flat-CE still leads macro-F1 on the gold distribution, quantifying the cost of ordinal modeling under label shift. Future work: Cohen's κ on the overlap set, a multi-LLM label model, soft ordinal targets, and a real gated encoder.

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- CORN/GCE/CORAL/Focal/Snorkel/Confident-Learning/Class-Balanced per-paper analyses: notes 45–51 in `docs/papers/finetuning-message/`.